

Superstore SQL Analysis

Customer table

select * from customers;

	Customer_ID	Customer_Name	Segment
1	AA-10315	Alex Avila	Consumer
2	AA-10375	Allen Arnold	Consumer
3	AA-10480	Andrew Allen	Consumer
4	AA-10645	Anna Andreadi	Consumer
5	AB-10015	Aaron Bergman	Consumer
6	AB-10060	Adam Bellavance	Home Office
7	AB-10105	Adrian Barton	Consumer
8	AB-10150	Aimee Bixby	Consumer
9	AB-10165	Alan Barnes	Consumer
10	AB-10255	Alejandro Ballentine	Home Office

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 793 rows

Products table

select * from products;

	Product_ID	Category	Sub_Category	Product_Name
1	FUR-BO-10000112	Furniture	Bookcases	Bush Birmingham Collection Bookcase, Dark Cherry
2	FUR-BO-10000330	Furniture	Bookcases	Sauder Camden County Barrister Bookcase, Planked ...
3	FUR-BO-10000362	Furniture	Bookcases	Sauder Inglewood Library Bookcases
4	FUR-BO-10000468	Furniture	Bookcases	O'Sullivan 2-Shelf Heavy-Duty Bookcases
5	FUR-BO-10000711	Furniture	Bookcases	Hon Metal Bookcases, Gray
6	FUR-BO-10000780	Furniture	Bookcases	O'Sullivan Plantations 2-Door Library in Landvery Oak
7	FUR-BO-10001337	Furniture	Bookcases	O'Sullivan Living Dimensions 2-Shelf Bookcases
8	FUR-BO-10001519	Furniture	Bookcases	O'Sullivan 3-Shelf Heavy-Duty Bookcases
9	FUR-BO-10001567	Furniture	Bookcases	Bush Westfield Collection Bookcases, Dark Cherry Fl...
10	FUR-BO-10001601	Furniture	Bookcases	Sauder Mission Library with Doors, Fruitwood Finish

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Orders table

Select * from orders;

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-BO-10001798	261.959991455078	2	0	41.9136009216309
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.940002441406	3	0	219.582000732422
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	OFF-LA-10000240	14.6199998855591	2	0	6.87139987945557
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	FUR-TA-10000577	957.577514648438	5	0.449999988079071	-383.031005859375
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	OFF-ST-10000760	22.3680000305176	2	0.200000002980232	2.51640009880066
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	FUR-FU-10001487	48.8600006103516	7	0	14.1694002151489
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	OFF-AR-10002833	7.28000020980835	4	0	1.96560001373291
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002275	907.151977539063	6	0.200000002980232	90.715202331543
9	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	OFF-BI-10003910	18.503999710083	3	0.200000002980232	5.78249979019165
10	10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	OFF-AP-10002892	114.900001525879	5	0	34.4700012207031

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--Step 2: Perform Required Queries

--Write and execute SQL queries for each of the following:

--1.) Find all orders where sales are greater than the average sales. (Subquery)

```
select *  
from orders  
where sales > (select avg(sales) from orders);
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-BO-10001798	261.959991455078	2	0	41.9136009216309
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.940002441406	3	0	219.582000732422
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	FUR-TA-10000577	957.577514648438	5	0.449999988079071	-383.031005859375
4	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002275	907.151977539063	6	0.200000002980232	90.715202331543
5	11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	FUR-TA-10001539	1706.18395996094	9	0.200000002980232	85.309196472168
6	12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002033	911.424011230469	4	0.200000002980232	68.3567962646484
7	14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	OFF-BH-10003656	407.876013183594	3	0.200000002980232	132.592193603516
8	17	CA-2014-105893	2014-11-11	2014-11-18	Standard Class	PK-19075	OFF-ST-10004106	665.880004882813	6	0	13.3176002502441
9	25	CA-2015-106320	2015-09-25	2015-09-30	Standard Class	EB-13870	FUR-TA-10000577	1044.63000488281	3	0	240.264892578125
10	28	US-2015-150630	2015-09-17	2015-09-21	Standard Class	TB-21520	FUR-BO-10004834	3083.42993164063	7	0.5	-1665.05224609375

Query executed successfully.

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--2.) Find the highest sales order for each customer. (Subquery)

```
select *  
from orders o1  
where Sales = (select max(sales) from orders o2 where o1.Customer_ID = o2.Customer_ID);
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.940002441406	3	0	219.582000732422
2	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	FUR-TA-10000577	957.577514648438	5	0.449999988079071	-383.031005859375
3	11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	FUR-TA-10001539	1706.18395996094	9	0.200000002980232	85.309196472168
4	25	CA-2015-106320	2015-09-25	2015-09-30	Standard Class	EB-13870	FUR-TA-10000577	1044.63000488281	3	0	240.264892578125
5	28	US-2015-150630	2015-09-17	2015-09-21	Standard Class	TB-21520	FUR-BO-10004834	3083.42993164063	7	0.5	-1665.05224609375
6	36	CA-2016-117590	2016-12-08	2016-12-10	First Class	QH-14485	TEC-PH-10004977	1097.5439453125	7	0.200000002980232	123.473701477051
7	39	CA-2015-117415	2015-12-27	2015-12-31	Standard Class	SN-20710	FUR-BO-10002545	532.399169921875	3	0.319999992847443	-46.9763984680176
8	55	CA-2016-105816	2016-12-11	2016-12-17	Standard Class	JM-15265	TEC-PH-10002447	1029.94995117188	5	0	298.685485839844
9	58	CA-2016-111682	2016-06-17	2016-06-18	First Class	TB-21055	FUR-CH-10003968	319.410003662109	5	0.100000001490116	7.09800004959106
10	68	CA-2014-106376	2014-12-05	2014-12-10	Standard Class	BS-11590	OFF-AR-10002671	1113.02404785156	8	0.200000002980232	111.302398681641

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 795 rows

--3.) Calculate total sales for each customer. (CTE)

```
with sum_sales as(  
    select customer_id,  
    sum(sales)
```

```

over (partition by customer_id) as total_sales

from orders)

select distinct customer_id,

total_sales

from sum_sales;

```

	customer_id	total_sales
1	FC-14245	3418.74196219444
2	SV-20935	1903.49098443985
3	GR-14560	1284.38000679016
4	EM-14140	4588.43589305878
5	JC-15340	71.2629990577698
6	MM-18055	2749.8810133934
7	MG-17650	3785.2759718895
8	JH-15910	959.476995944977
9	BK-11260	659.138000011444
10	NF-18475	377.162011146545

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 793 rows

--4.) Find customers whose total sales are above average. (CTE + Subquery)

```

with sum_sales as(

select customer_id,

sum(sales) as total_sales

from orders

group by customer_id

)

select *

from sum_sales

where total_sales > (select avg(total_sales)

from sum_sales);

```

	customer_id	total_sales
1	LF-17185	3930.50898265839
2	KH-16330	3312.86404490471
3	SC-20305	3979.06489086151
4	SV-20785	4105.30506920815
5	JL-15505	3173.87399768829
6	RB-19360	15117.3389587402
7	FP-14320	4046.74587917328
8	HR-14830	5248.78697681427
9	JM-15250	3159.12198781967
10	SR-20740	4345.8859823547

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 294 rows

--5.) Rank all customers based on total sales. (Window Function)

```

with sum_sales as(

```

```

select customer_id,

sum(sales) as total_sales

from orders

group by customer_id)

select customer_id,

total_sales,

rank() over (order by total_sales desc)

from sum_sales;

```

	customer_id	total_sales	(No column name)
1	SM-20320	25043.0504960878	1
2	TC-20980	19052.217195034	2
3	RB-19360	15117.3389587402	3
4	TA-21385	14595.6198644638	4
5	AB-10105	14473.571243763	5
6	KL-16645	14175.2289050817	6
7	SC-20095	14142.3341379166	7
8	HL-15040	12873.297662735	8
9	SE-20110	12209.4381403923	9
10	CC-12370	12129.0715373755	10

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 793 rows

--6.) Assign row numbers to each order within a customer. (Window Function + PARTITION BY)

```

select *,

row_number() over(partition by customer_id order by order_date) as order_number

from orders;

```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit	order_number
1	2230	CA-2014-128055	2014-03-31	2014-04-05	Standard Class	AA-10315	OFF-BL-10004390	673.567993164063	2	0.200000002980232	252.587997436523	1
2	2231	CA-2014-128055	2014-03-31	2014-04-05	Standard Class	AA-10315	OFF-AP-10002765	52.9799995422363	2	0	14.834400177002	2
3	7469	CA-2014-138100	2014-09-15	2014-09-20	Standard Class	AA-10315	OFF-PA-10000349	14.9399995803833	3	0	7.02180004119873	3
4	7470	CA-2014-138100	2014-09-15	2014-09-20	Standard Class	AA-10315	FUR-FU-10002456	14.5600004196167	2	0	6.26079988479614	4
5	1300	CA-2015-121391	2015-10-04	2015-10-07	First Class	AA-10315	OFF-ST-10001590	26.9599990844727	2	0	7.0096001625061	5
6	5199	CA-2016-103982	2016-03-03	2016-03-08	Standard Class	AA-10315	OFF-SU-10000151	3930.07202148438	3	0.200000002980232	-786.014404296875	6
7	5200	CA-2016-103982	2016-03-03	2016-03-08	Standard Class	AA-10315	OFF-FA-10001332	2.30399990081787	1	0.200000002980232	0.777599990367889	7
8	5201	CA-2016-103982	2016-03-03	2016-03-08	Standard Class	AA-10315	TEC-PH-10000895	431.976013183594	3	0.200000002980232	32.3982009887695	8
9	5202	CA-2016-103982	2016-03-03	2016-03-08	Standard Class	AA-10315	TEC-AC-10002857	41.7200012207031	7	0.200000002980232	5.73649978637695	9
10	1160	CA-2017-147039	2017-06-29	2017-07-04	Standard Class	AA-10315	OFF-AP-10000576	362.940002441406	3	0	90.7350006103516	10

Query executed successfully.localhost\SQLEXPRESS (17.0...LAPTOP-H2VG5D2U\user (56)superstoreDataset00:00:00Row: 1, Col: 19,994 rows

--7.) Display top 3 customers based on total sales. (Window Function)

```

with sum_sales as(

select customer_id,

sum(sales) as total_sales

from orders

group by customer_id

```

```

),
rnk_sales as(
    select customer_id,
    total_sales,
    rank() over (order by total_sales desc) as rnk
    from sum_sales
)
select *
from rnk_sales
where rnk <= 3;

```

	customer_id	total_sales	rnk
1	SM-20320	25043.0504860878	1
2	TC-20980	19052.217195034	2
3	RB-19360	15117.3389587402	3

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | LAPTOP-H2VG5D2U\user (56) | superstoreDataset | 00:00:00 | Row: 1, Col: 1 | 3 rows

--Step 3: Final Combined Query

--Write one final query that shows:

--Customer Name

--Total Sales

--Rank

--(Use JOIN + CTE + Window Function together)

```

with sum_sales as(
    select c.customer_name,
    o.customer_id,
    sum(o.sales) as total_sales
    from customers c

```

```

join orders o
on c.customer_id = o.customer_id
group by o.customer_id,c.customer_name
)
select customer_name,
total_sales,
rank() over (order by total_sales) as rnk
from sum_sales;

```

	customer_name	total_sales	rnk
1	Thais Sissman	4.83300018310547	1
2	Lela Donovan	5.30399990081787	2
3	Carl Jackson	16.5200004577637	3
4	Mitch Gastineau	16.7389998435974	4
5	Roy Skaria	22.3280000686646	5
6	Susan Gilcrest	47.946001291275	6
7	Ricardo Emerson	48.3600006103516	7
8	Larry Blacks	50.1879988908768	8
9	Adrian Shami	58.8199982643127	9
10	Jasper Cacioppo	71.2629990577698	10

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | LAPTOP-H2VG5D2U\user (56) | superstoreDataset | 00:00:00 | Row: 1, Col: 1 | 793 rows

--Mini Project: Customer Sales Insights

--Answer the following using SQL:

--1.) Who are the top 5 customers?

```

with sum_sales as(
select c.customer_name,
o.customer_id,
sum(o.sales) as total_sales
from orders o
join customers c
on o.customer_id = c.customer_id
group by c.customer_name,o.customer_id
),
rnk_sales as(
select customer_name,

```

```

customer_id,
total_sales,
rank() over (order by total_sales desc) as rnk
from sum_sales
)
select customer_name,
customer_id,
total_sales
from rnk_sales
where rnk <= 5;

```

	customer_name	customer_id	total_sales
1	Sean Miller	SM-20320	25043.0504860878
2	Tamara Chand	TC-20980	19052.217195034
3	Raymond Buch	RB-19360	15117.3389587402
4	Tom Ashbrook	TA-21385	14595.6198644638
5	Adrian Barton	AB-10105	14473.571243763

Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset | 00:00:00 | Row: 1, Col: 1 | 5 rows

--2.) Who are the bottom 5 customers?

```

with sum_sales as(
    select c.customer_name,
           o.customer_id,
           sum(o.sales) as total_sales
    from orders o
    join customers c
    on o.customer_id = c.customer_id
    group by c.customer_name,o.customer_id
),
rnk_sales as(
    select customer_name,
           customer_id,
           total_sales,

```

```

rank() over (order by total_sales asc) as rnk
from sum_sales
)
select customer_name,
customer_id,
total_sales
from rnk_sales
where rnk <= 5;

```

	customer_name	customer_id	total_sales
1	Thais Sissman	TS-21085	4.83300018310547
2	Lela Donovan	LD-16855	5.30399990081787
3	Carl Jackson	CJ-11875	16.5200004577637
4	Mitch Gastineau	MG-18205	16.7389998435974
5	Roy Skaria	RS-19870	22.3280000686646

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | LAPTOP-H2VG5D2U\user (56) | superstoreDataset | 00:00:00 | Row: 1, Col: 1 | 5 rows

--3.) Which customers made only one order?

```

select c.customer_name,
o.customer_id,
count(distinct o.order_id) as order_count
from orders o
join customers c
on c.customer_id = o.customer_id
group by c.customer_name,o.customer_id
having count(distinct o.order_id) = 1;

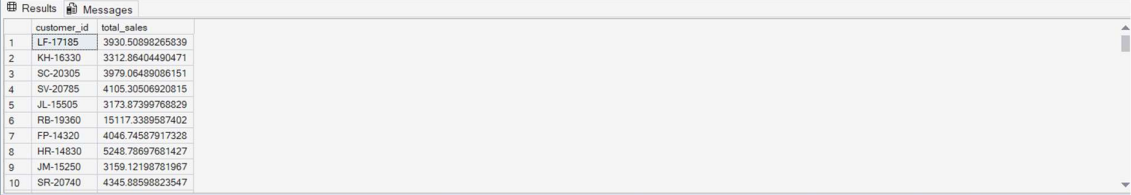
```

	customer_name	customer_id	order_count
1	Anthony O'Donnell	AO-10810	1
2	Anemone Ratner	AR-10570	1
3	Carl Jackson	CJ-11875	1
4	Jenna Calley	JC-15385	1
5	Jocasta Rupert	JR-15700	1
6	Lela Donovan	LD-16855	1
7	Mitch Gastineau	MG-18205	1
8	Patricia Hirasaki	PH-18790	1
9	Ricardo Emerson	RE-19405	1
10	Roland Murray	RM-19750	1

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | LAPTOP-H2VG5D2U\user (56) | superstoreDataset | 00:00:00 | Row: 1, Col: 1 | 10 rows

--4.) Which customers have above-average sales?

```
with sum_sales as(
    select customer_id,
           sum(sales) as total_sales
    from orders
    group by customer_id
)
select *
from sum_sales
where total_sales > (select avg(total_sales)
                    from sum_sales);
```



The screenshot shows a SQL query results window with a table containing 10 rows. The columns are 'customer_id' and 'total_sales'. The data is as follows:

	customer_id	total_sales
1	LF-17185	3930.50898265839
2	KH-16330	3312.86404490471
3	SC-20305	3979.06489086151
4	SV-20785	4105.30506920815
5	JL-15505	3173.87399768829
6	RB-19360	15117.3389587402
7	FP-14320	4046.74587917328
8	HR-14830	5248.78697681427
9	JM-15250	3159.12198781967
10	SR-20740	4345.88598823547

Below the table, a status bar indicates: 'Query executed successfully. localhost\SQLEXPRESS (17.0 ... LAPTOP-H2VG5D2U\user (56) superstoreDataset 00:00:00 Row: 1, Col: 1 294 rows'.

--5.) What is the highest order value per customer?

```
select o.customer_id,
       c.customer_name,
       max(o.sales) as highest_order
from orders o
join customers c
on c.customer_id = o.customer_id
group by o.customer_id, c.customer_name
order by highest_order desc;
```

ResultsMessages

	customer_id	customer_name	highest_order
1	SM-20320	Sean Miller	22638.48046875
2	TC-20980	Tamara Chand	17499.94921875
3	RB-19360	Raymond Buch	13999.959609375
4	TA-21385	Tom Ashbrook	11199.9677734375
5	HL-15040	Hunter Lopez	10499.9697265625
6	AB-10105	Adrian Barton	9892.740234375
7	SC-20095	Sanjit Chand	9449.9501953125
8	BS-11365	Bill Shonely	9099.9296875
9	SE-20110	Sanjit Engle	8749.9501953125
10	CC-12370	Christopher Conant	8399.9755859375

Query executed successfully.localhost\SQLEXPRESS (17.0 ...LAPTOP-H2VG5D2U\user (56)superstoreDataset00:00:00Row: 1, Col: 1793 rows